OCNMS Dive Survey Annual Report

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Introduction

Divers from the Northwest Fisheries Science Center (NWFSC) and the National Marine Sanctuaries (NMS) have conducted kelp-forest dive surveys in Olympic Coast National Marine Sanctuary (OCNMS) since 2015 (with the exception of 2020 due to COVID restrictions). Divers from the University of Washington and California State University Monterey Bay have participated as reciprocity divers. Surveys are completed at the end of July to beginning of August. In 2026, surveys were competed between 2026-07-30 and 2026-08-04.

In 2026, divers completed 72 fish transects, 60 invertebrate transects, and 60 across five (5) sites (Fig. 1). After filtering the data for visibility (>2m), there were 64 usable fish transects in 2026. Divers recorded 23, 47, and 14 taxa of fish, invertebrates, and algae along these transects. To date (2015-2026), the survey has recorded 41, 76, and 17 taxa for fish, invertebrates, and algae.

Methods

Divers on SCUBA conducted in situ surveys to count fish at each site along benthic belt transects (30 m by 2 m) following procedures modified from @malone2022large. Transects were conducted within or directly adjacent to canopy kelp beds (consisting of giant Macrocystis pyrifera or bull Nereocystis luetkeana kelps). In 2015 surveyed at 10 sites and conducted four (4) transects per site at 5 m depth (Fig. 1, [@shelton2018predictable]). From 2016 on, we surveyed at five (5) sites (Fig. 1), sampling at two (2) locations within each site separated by >100 m, and 2 depths within each location (5 and 10 m) Our goal was to complete six (6) replicate transects at each year-site-depth combination [@tolimieri2023changes].

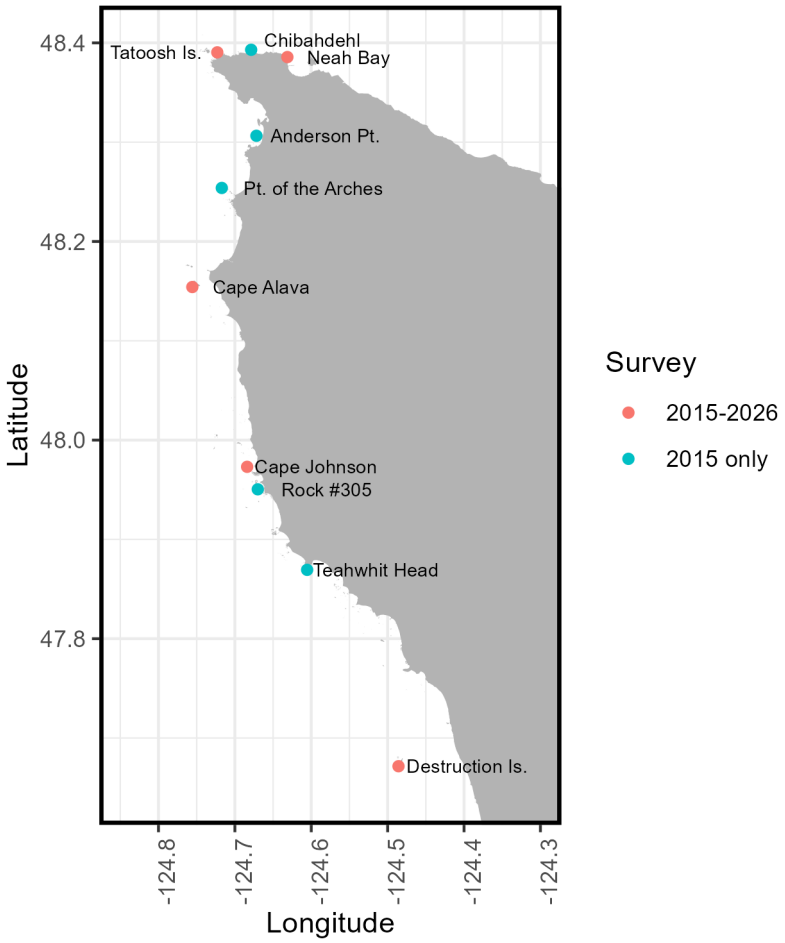


Figure 1: Dive survey locations in Olympic Coast National Marine Sanctuary along the coast of Washington State.

During each fish transect, we counted and estimated the size (total length to nearest cm) of all fishes >5 cm total length; the exception was rockfishes Sebastes spp., for which we estimated sizes of all individuals. Rockfishes $\leq$ 10 cm were considered young-of-year. Divers also estimated horizontal visibility on each transect by determining the distance at which the lead diver could distinguish their buddy's extended fingers. Transects with visibility less than 2 m were excluded from analyses.

Divers / Data collectors

Many divers have contributed to the survey over the years. They include:

Dive team in 2026: Jameal Samhouri, Greg Williams, Nick Tolimieri, Alison Haupt, Kinsey Frick, Megan Shaffer, and Ole Shelton

Dive team members: Greg Williams, Steve Lonhart, Kelly Andrews, Nick Tolimieri, Jameal Samhouri, Joe Mangiafico, Gary Longo, Genoa Sullaway, Ole Shelton, Chris Harvey, Kinsey Frick, Paul Chittaro, Emily Bishop, Megan Shaffer, Alison Haupt, and Catalina Burch

Results or some other title

Fish

need to fix table referencing; insert fake table?

Divers target 12 transects per site (3 transects x 2 locations x 2 depths), but are often able to complete additional transects on a dive (Table 1). In 2026 divers completed 13-16 for a total of 72 transects across all sites. Note, the analysts typically remove transects with visibility below 2.0 m for any investigation.

why don't we have fish transects at Destruction Island in 2015

Table 1: Number of fish transects completed by year across 10 sites in 2015 and five sites from 2016 to the current year. Note, the data have not been filtered for visibility.

Site	2015	2016	2017	2018	2019	2021	2022	2023	2024	2025	2026
Teahwhit Head	4										
Cape Johnson	4	12	16	15	15	16	4	17	12	16	15
Rock 305	4										
Cape Alava	4	12	19	16	16	14	15	16	15	15	13
Point of the Arches	4										
Anderson Point	4										
Tatoosh Island	4	8	13	15	14	14	14	16	17	14	16
Chibadehl Rocks	4										
Neah Bay	4	10	16	15	15	16	13	17	17	16	15
Destruction Island		10	14	15	16	14	4	15	14	15	13

Table 2 shows the number of species observed at completed at each site. In 2026 divers recorded between 9-18. The highest number of species was observed at 5.

Species/taxa observed in 2026: c(2015, 2015, 2015, 2015, 2015, 2015, 2015, 2015, 2015, 2015, 2015, 2015, 2015, 2015, 2015, 2015, 2016, 2016, 2016, 2016, 2016, 2016, 2016, 2016, 2016, 2016, 2016, 2016, 2016, 2016, 2017, 2017, 2017, 2017, 2017, 2017, 2017, 2017, 2017, 2017, 2017, 2017, 2017, 2017, 2017, 2017, 2018, 2018, 2018, 2018, 2018, 2018, 2018, 2018, 2018, 2018, 2018, 2018, 2018, 2018, 2018, 2018, 2018, 2018, 2019, 2019, 2019, 2019, 2019, 2019, 2019, 2019, 2019, 2019, 2019, 2019, 2019, 2019, 2019, 2019, 2019, 2019, 2021, 2021, 2021, 2021, 2021, 2021, 2021, 2021, 2021, 2021, 2021, 2021, 2021, 2021, 2021, 2021, 2021, 2021, 2022, 2022, 2022, 2022, 2022, 2022, 2022, 2022, 2022, 2022, 2022, 2022, 2022, 2022, 2023, 2023, 2023, 2023, 2023, 2023, 2023, 2023, 2023, 2023, 2023, 2023, 2023, 2023, 2023, 2023, 2023, 2024, 2024, 2024, 2024, 2024, 2024, 2024, 2024, 2024, 2024, 2024, 2024, 2024, 2024, 2024, 2024, 2024, 2025, 2026), c("AUFL", "COTT", "EMBI", "EMLA", "ENBI", "HEDE", "HEHE", "HELA", "MYOP", "OPEL", "OXPI", "RHNI", "SCMA", "SECA", "SEMA", "SEME", "AUFL", "CHNU", "COTT", "EMLA", "HEDE", "HEHE", "HELA", "HEST", "OPEL", "OXPI", "RHNI", "RHVA", "SCMA", "SEME", "AUFL", "EMBI", "EMLA", "ENBI", "HEDE", "HELA", "OPEL", "OXPI", "RHNI", "RHVA", "RIMU", "SCMA", "SECA", "SEMA", "SEME", "SENE", "ARHA", "AUFL", "CHNU", "COTT", "EMLA", "HEDE", "HELA", "HEST", "JOZO", "OPEL", "RHNI", "RHVA", "SCMA", "SECA", "SEME", "SENE", "SYGI", "AUFL", "BAITBALL", "CLUP", "COTT", "EMLA", "HEDE", "HEHE", "HELA", "HEST", "OPEL", "OXPI", "RHNI", "RHVA", "RIMU", "SCMA", "SECA",

“SEMA”, “SEME”, “SENE”, “ARHA”, “AUFL”, “COTT”, “EMLA”, “HEDE”, “HEHE”, “HEST”, “OPEL”, “OXPI”, “PHOL”, “RHNI”, “SCMA”, “SECA”, “SEFL”, “SEME”, “SENE”, “SEPI”, “UNID”, “AUFL”, “CLUP”, “EMLA”, “HEDE”, “HEHE”, “OPEL”, “RHNI”, “SCMA”, “SEBYT”, “SECA”, “SEME”, “SENE”, “AMHE”, “AUFL”, “CITH”, “COTT”, “EMLA”, “HEDE”, “HEHE”, “HEXA”, “OPEL”, “OXPI”, “RHVA”, “RIMU”, “SCMA”, “SEMA”, “SEME”, “SENE”, “SEPI”, “AUFL”, “COTT”, “EMLA”, “ENBI”, “HEDE”, “HEHE”, “OPEL”, “OXPI”, “RHNI”, “RHVA”, “SCMA”, “SECA”, “SEMA”, “SEME”, “SEMY”, “SENE”, “SYGI”, “AUFL”, “COTT”, “EMLA”, “HEDE”, “HEHE”, “HELA”, “HEST”, “JOZO”, “OPEL”, “OXPI”, “PLCO”, “RHNI”, “RHVA”, “SCMA”, “SECA”, “SEFL”, “SEME”, “SEMY”, “SENE”, “SYGI”, “AUFL”, “BRFR”, “COTT”, “EMBI”, “EMLA”, “HEDE”, “HEHE”, “HELA”, “HEST”, “OPEL”, “OXPI”, “PHOL”, “RHVA”, “RIMU”, “SCMA”, “SECA”, “SEMA”, “SEME”, “SEMY”, “SENE”), c(43, 3, 9, 21, 1, 16, 2, 3, 1, 2, 1, 3, 1, 1, 17, 46, 106, 1, 1, 83, 57, 1, 1, 1, 6, 1, 5, 3, 2, 102, 45, 2, 126, 1, 79, 1, 8, 4, 3, 3, 2, 5, 5, 1, 371, 2, 14, 2, 2, 1, 84, 88, 1, 1, 4, 8, 10, 3, 7, 2, 225, 1, 5, 25, 200, 148, 4, 174, 176, 3, 1, 1, 13, 2, 5, 2, 1, 12, 8, 1, 532, 12, 1, 36, 2, 52, 145, 1, 2, 12, 2, 1, 2, 2, 4, 2, 199, 4, 2, 1, 90, 150, 26, 73, 1, 4, 3, 3, 11, 1, 141, 3, 1, 333, 2, 1, 110, 141, 3, 2, 7, 1, 7, 1, 8, 4, 420, 8, 2, 3, 4, 30, 1, 131, 2, 12, 1, 1, 4, 5, 5, 2, 284, 1, 8, 1, 172, 1, 227, 259, 3, 4, 10, 2, 18, 5, 1, 5, 9, 7, 4, 12, 467, 7, 1, 1, 70, 1, 1, 1, 149, 171, 2, 14, 4, 13, 12, 1, 16, 2, 6, 2, 1, 353, 10, 2)
Total species/taxa observed to date: 35

Table 2: Number of fish taxa observed by year across 10 sites.

Site	2015	2016	2017	2018	2019	2021	2022	2023	2024	2025	2026
Teahwhit Head	6										
Rock 305	1										
Cape Alava	5	11	10	10	10	11	5	14	12	17	18
Point of the Arches	6										
Anderson Point	1										
Tatoosh Island	4	9	6	10	13	6	5	6	7	9	11
Chibadehl Rocks	3										
Neah Bay	9	10	11	11	16	12	10	12	18	15	14
Destruction Island		6	11	9	12	16		11	11	13	12
Cape Johnson		7	8	9	11	12		10		11	9

Adult fishes

Across all years, divers recorded 37 adult fish taxa (Table 3). The most common species was black rockfish Sebastes melanops with 3140 counted to date. The second most common species was kelp greenling HEDE 1336individuals observed by divers. In 2026 black rockfish was`text1`the most commonly recorded species with353 across the five sites.

Table 3

Species	Common name	Total
Sebastes melanops	black rockfish	3,140
Hexagrammos decagrammus	kelp greenling	1,336
Embiotoca lateralis	striped surfperch	1,082
Aulorhynchus flavidus	tubesnout	925

Species	Common name	Total
Clupeidae	herring	298
	bait-sardines-anchovy	200
Ophiodon elongatus	lingcod	103
Scorpaenichthys marmoratus	cabezon	58
Rhacochilus vacca	pile perch	47
Sebastes nebulosus	China rockfish	41
Rhinogobiops nicholsii	blackeye goby	37
Sebastes caurinus	copper rockfish	32
Oxylebius pictus	painted greenling	29
Sebastes maliger	quillback rockfish	26
Hexagrammos lagocephalus	rock greenling	25
Hexagrammos stelleri	whitespotted greenling	19
Cottidae	sculpins	18
Hemilepidotus hemilepidotus	red Irish lord	18
Sebastes mystinus	blue rockfish	18
Artedius harringtoni	scalyhead sculpin	15
Sebastes flavidus	yellowtail rockfish	14
Embiotocidae	surfperches	12
Sebastes melaanops-flavidus YOY	yellowtail-black YOY	11
Synchirus gilli	manacled sculpin	7
Jordania zonope	longfin sculpin	6
Rimicola muscarum	kelp clingfish	6
Sebastes pinniger	canary rockfish	4
Chirolophis nugator	mosshead warbonnet	3
Enophrys bison	buffalo sculpin	3
Citharichthys spp	sanddabs	2
Hexagrammos spp	greenlings	2
Pholididae	gunnels	2
Ammodytes personatus	Pacific sand lance	1
Brachyistius frenatus	kelp surfperch	1
Myoxocephalus polyacanthocephalus	great sculpin	1
Pleuronichthys coenosus	c-o sole	1
unidentified fish	unidentified fish	1

As noted black rockfish (Sebastes melanostictus) is the most commonly observed species in the dive survey

(Figure 2). The abundance of black rockfish decreased from 6.8 (± 1.1 se) 120 m ^{3} in 2025 to 4.9 (± 0.7 se) 120 m ^{3} in 2026. Greenlings are typically the second most observed taxa; these are primarily kelp greenlings Hexagrammos decagrammus, although the various species are lumped in Figure 2.

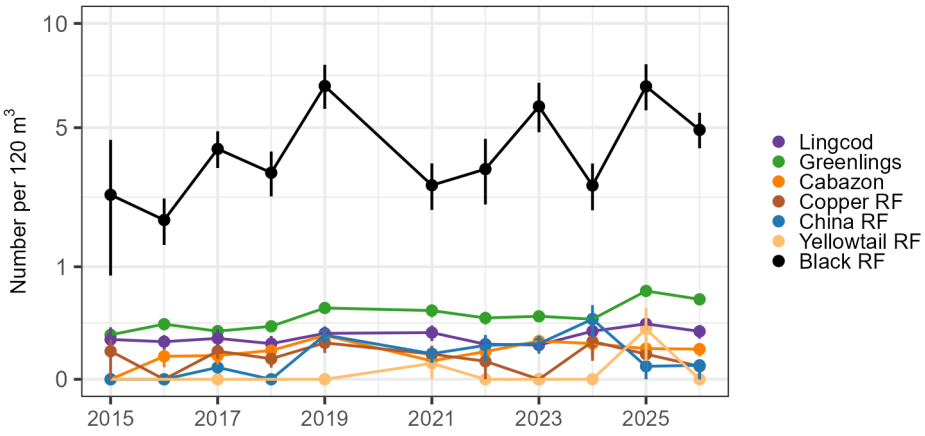


Figure 2: Number of fish per 120 m^3 for 2015-2026 coastwide. Note the scale of the y-axis. RF = rockfish.

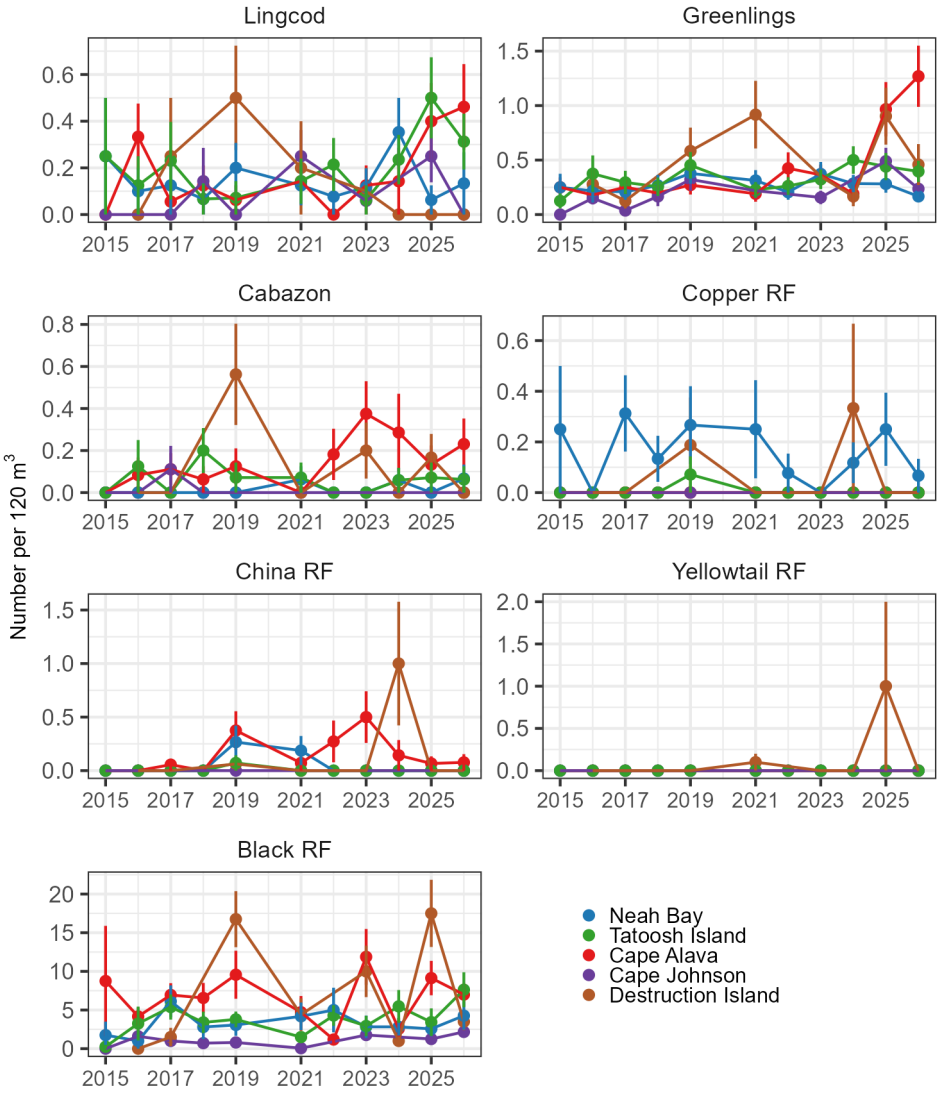


Figure 3: Number of fish per 120 m^3 for 2015-2026 by site.

Rockfish young-of-year

It is difficult to visually distinguish many rockfish species when they are small. Therefore, on our surveys, we categorized juvenile rockfishes into five (5) groups established in the literature [@johansson2018seasonal; @markel2020contrasting]:

(1) Yellowtail and black (YTB) included both yellowtail (S. flavidus) and black (S. melanops) rockfishes

(2) The copper/quillback/brown (CQB) group included copper (S. caurinus), quillback (S. maliger), and brown (S. auriculatus) rockfishes

(3) Canary (S. pinniger)

(4) Blue rockfish (S. mystinus)

(5) Unidentified individuals were categorized as juvenile rockfishes

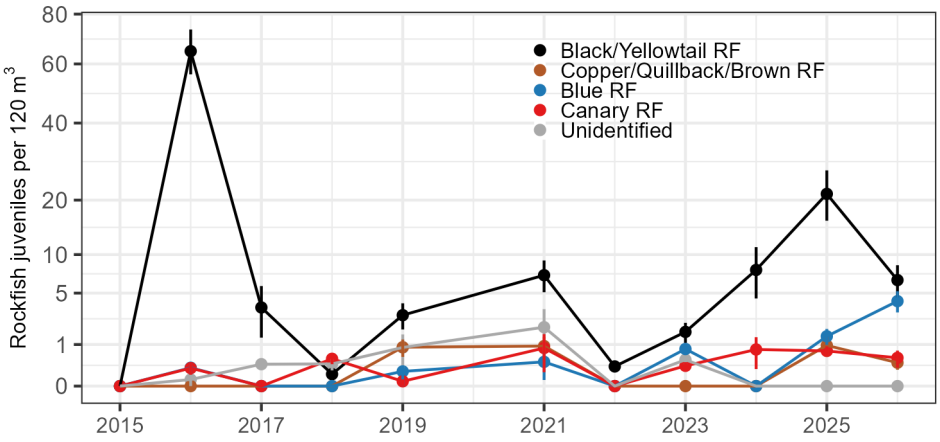


Figure 4: Number of rockfish young-of-year from 2015-2026 coastwide.

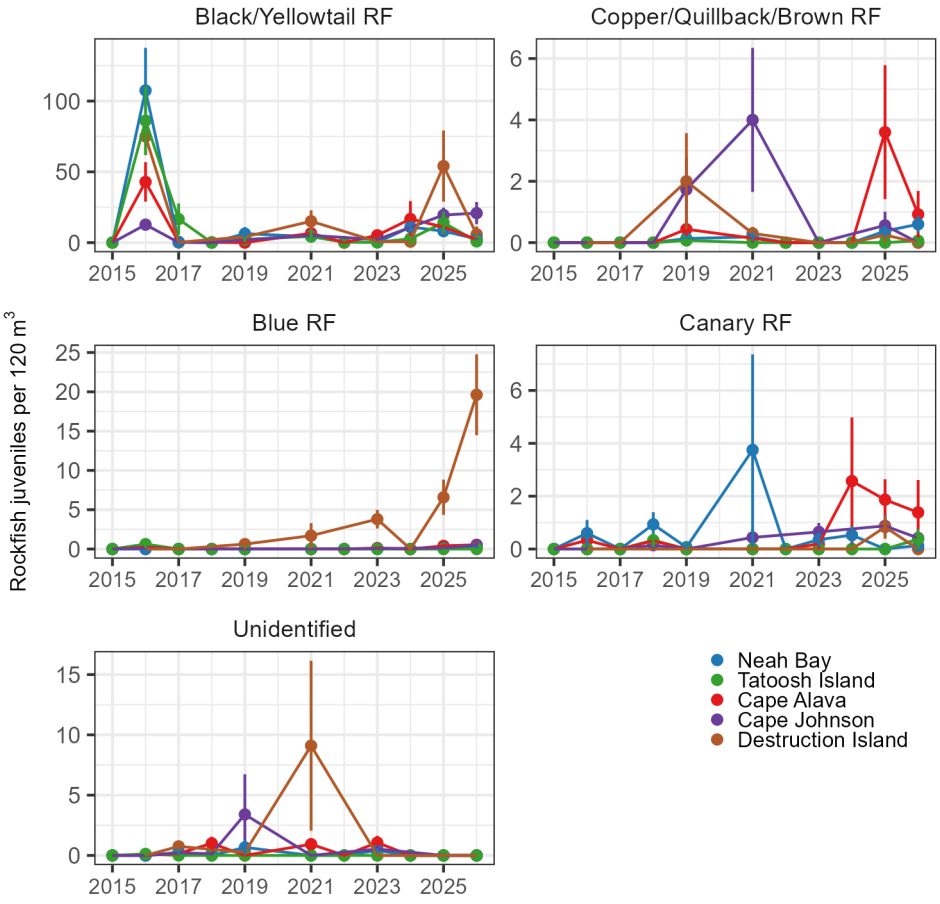


Figure 5: Number of rockfish young-of-year from 2015-2026 coastwide.

Invertebrates

Table 4: Number of kelp transects completed by year across 10 sites in 2015 and five sites from 2016 to the current year.

Site	2015	2016	2017	2018	2019	2021	2022	2023	2024	2025	2026
Neah Bay	4	10	12	16	14	13	11	15	13	13	11
Tatoosh Island	4	9	11	12	9	13	13	12	18	11	10
Cape Alava	4	10	14	12	14	12	13	15	13	13	14
Cape Johnson	3	13	13	12	14	14	12	16	13	19	13
Destruction Island	2	7	12	14	11	11	12	12	12	14	12

Urchin trends

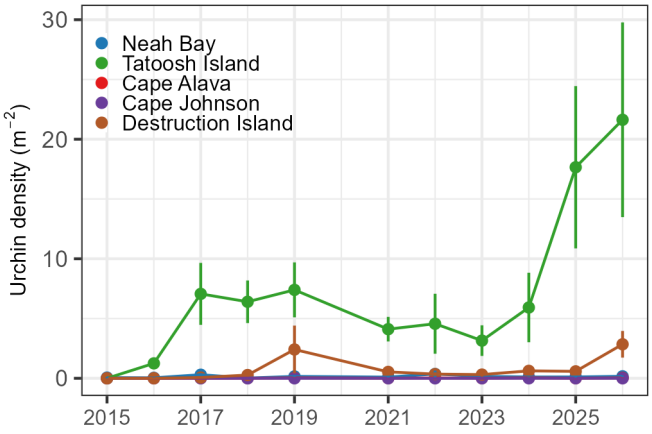


Figure 6

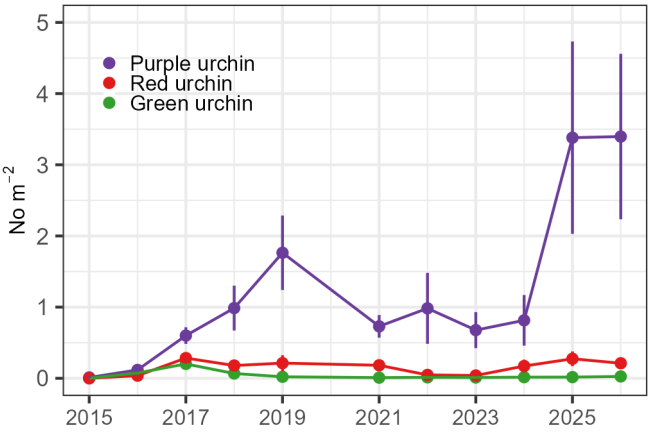


Figure 7

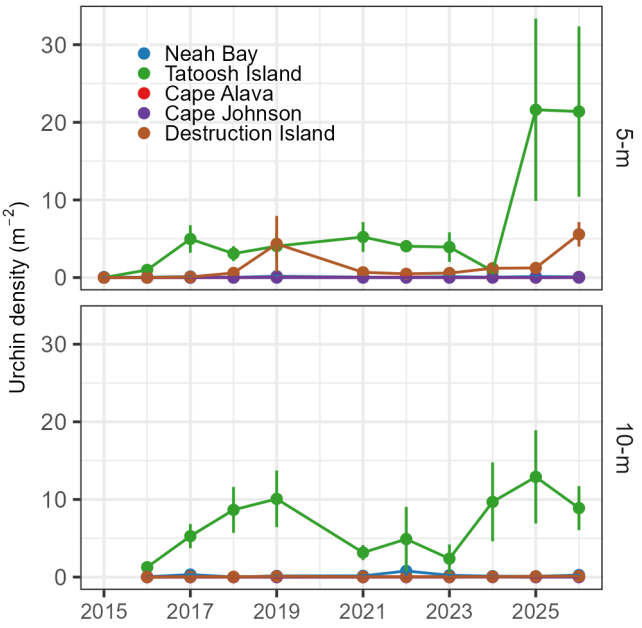


Figure 8

Seastars

All species

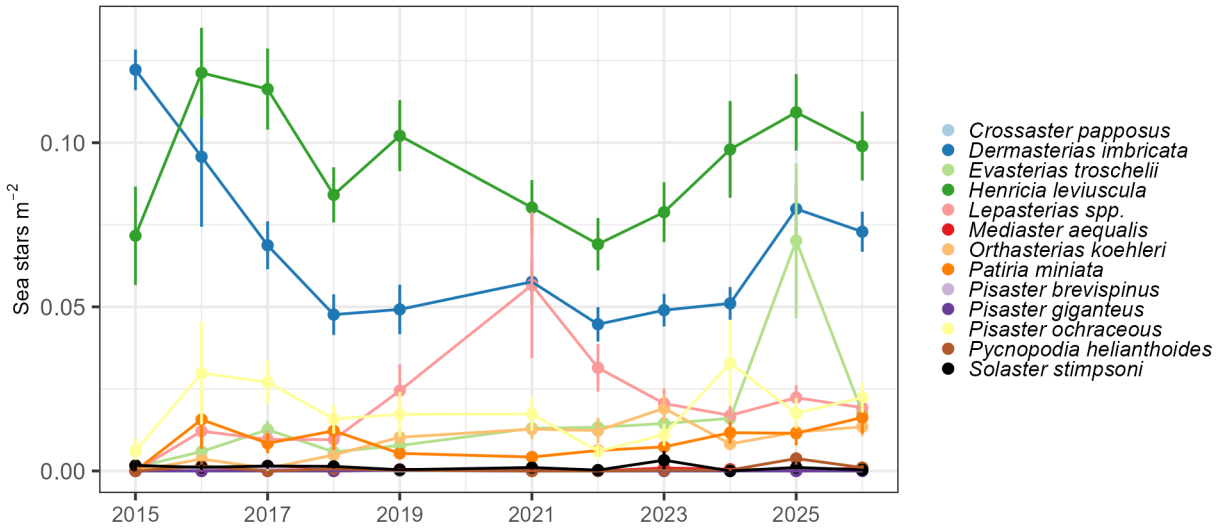


Figure 9

Pycnopodia

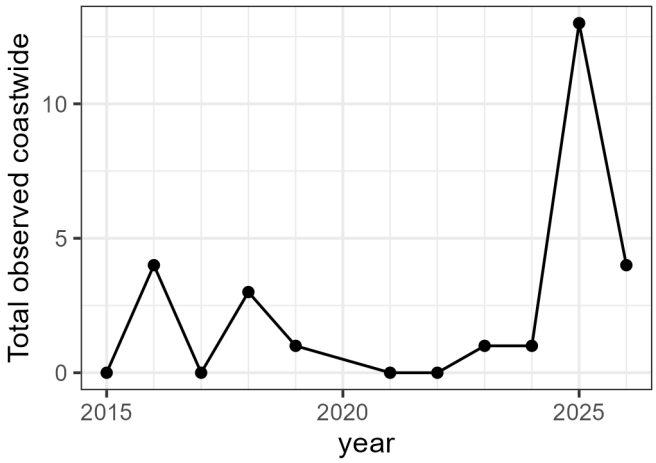


Figure 10

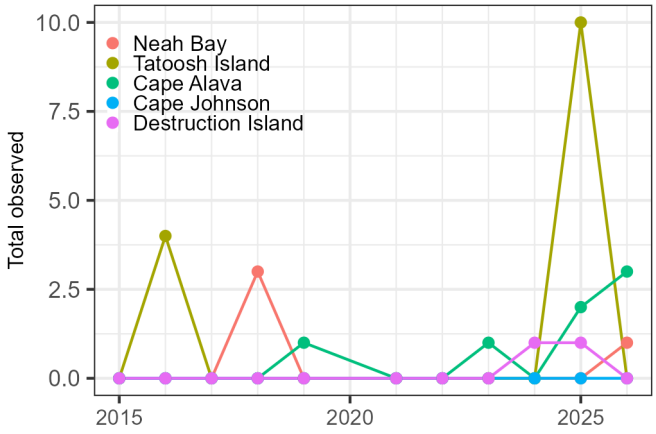


Figure 11

Algae

Table 5: Number of kelp transects completed by year across 10 sites in 2015 and five sites from 2016 to the current year.

Site	2015	2016	2017	2018	2019	2021	2022	2023	2024	2025	2026
Neah Bay	4	11	13	14	14	13	11	13	14	14	12
Tatoosh Island	6	11	11	12	9	13	11	12	12	13	10
Cape Alava	6	11	14	13	14	13	13	16	14	14	13
Cape Johnson	3	13	13	12	14	14	12	16	13	13	13
Destruction Island	3	8	13	16	11	12	12	12	10	15	12

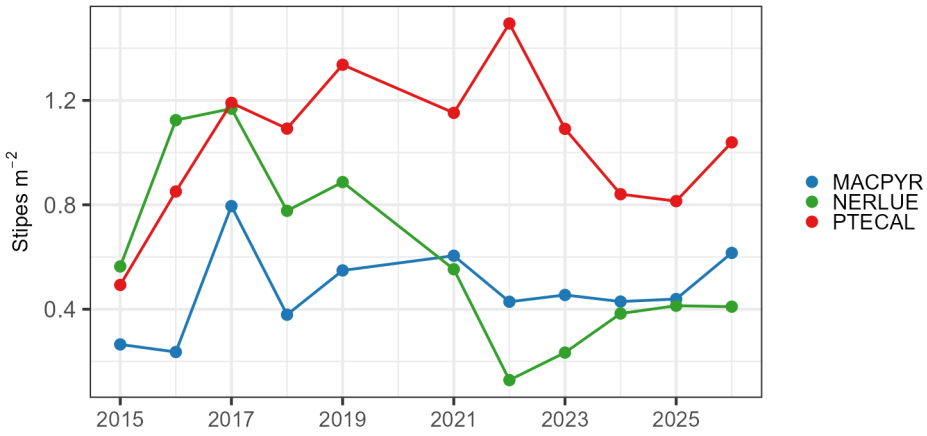


Figure 12

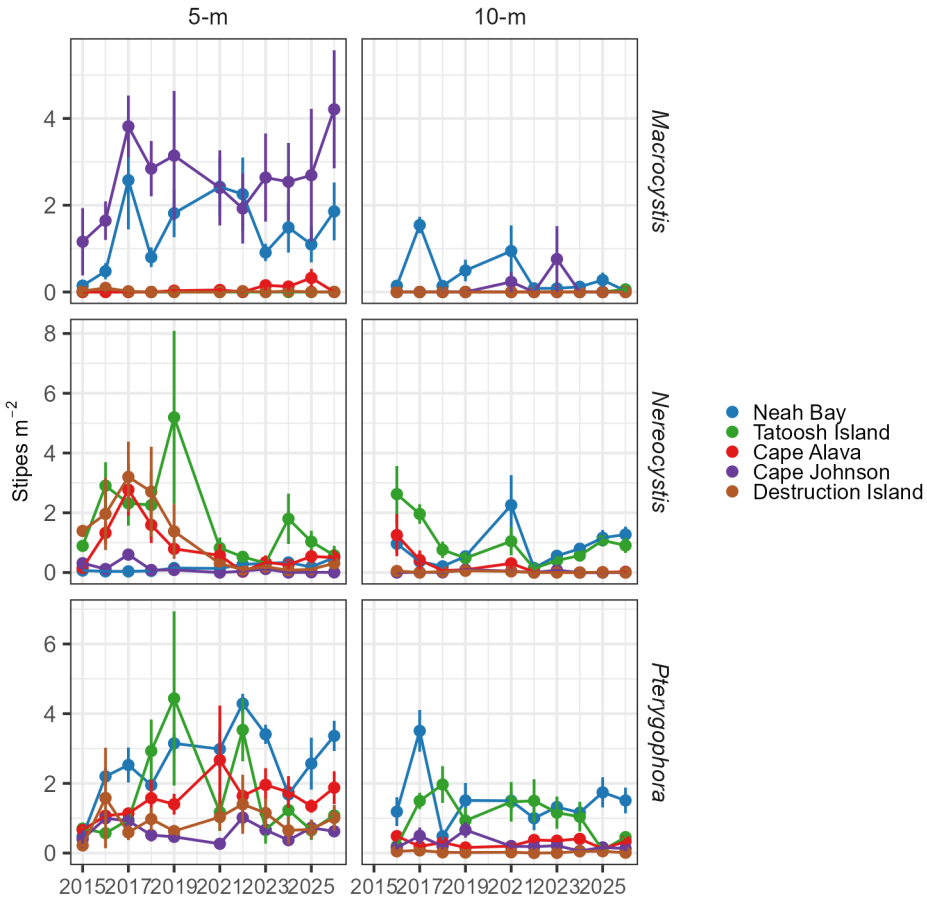


Figure 13

Discussion

Literature Cited